

SOFTWARE REQUIREMENTS SPECIFICATION

SkillBridge

Skill-Based Job Matching & Career Guidance Portal

From skill assessment to career clarity — powered by real skill data, not guesswork.

Document Version: 2.0 (Full-Stack Build)

Tech Stack: React.js + Tailwind CSS | Python | MongoDB

Prepared for: Hackathon / Academic Evaluation

Table of Contents

1. Introduction

1.1 Purpose

SkillBridge is a full-stack, skill-based career guidance portal built for pre-final year students. It replaces generic job-listing platforms with a system that measures a student's actual skills, matches them against real job-role requirements, and shows exactly what is missing to become job-ready — backed by a live React + Python + MongoDB application rather than a static prototype.

1.2 Problem Statement

Many students apply for jobs and internships without knowing whether they actually meet the skill requirements. Existing job portals list openings but stop there — they do not tell a student what skills they are missing, how big the gap is, or what to do about it. SkillBridge closes that gap by analyzing skills first and turning the result into personalized, actionable career guidance.

1.3 Scope

- Structured skill self-assessment with proficiency sliders / questionnaire
- Career matching engine that computes a compatibility score per job role
- Skill-gap identification and visualization against any target role
- Personalized course recommendations mapped to each skill gap
- Counselor dashboard for profile review, guidance notes, and cohort insights
- Skill growth tracking across semesters

1.4 Intended Users

The portal is built around four user roles, each with a distinct purpose:

User Role	What They Do on SkillBridge
Students	Take the skill assessment, view career matches, explore skill gaps, and follow course recommendations.
Career Counselors	Review student skill profiles, monitor progress, and log personalized guidance notes.

User Role	What They Do on SkillBridge
Recruiters	View verified, skill-based candidate profiles instead of resumes alone.
Training Providers	Get visibility into common skill gaps and align course offerings to real demand.

1.5 Platform

SkillBridge is a responsive, full-stack web application — a React.js + Tailwind CSS frontend backed by a Python API layer and a MongoDB database — usable on laptops, tablets, and mobile devices, including during on-campus placement drives.

2. Why SkillBridge Is Different

Generic job portals and career-advice sites stop at listing openings or offering generic tips; they never tie recommendations back to a student's measured skills or track change over time. SkillBridge is unique in closing that loop end-to-end.

Aspect	Existing Job Portals	SkillBridge
Job discovery	Shows available jobs	Matches jobs based on actual measured skills
Skill gap analysis	Not available	Identifies and visualizes missing skills per role
Learning guidance	No personalized learning path	Recommends specific courses to close each gap
Progress tracking	No progress tracking	Tracks skill growth over time / across semesters
Human guidance	No counselor support	Dedicated counselor dashboard with logged notes
Candidate insight	Resume keywords only	Skill-verified profiles recruiters can trust

3. System Overview

SkillBridge guides a student through a structured self-assessment, builds a live skill profile, and continuously compares that profile against a catalog of job roles stored in MongoDB. The result is a ranked list of career matches, a visual skill-gap breakdown for any chosen role, and course recommendations pulled dynamically from the backend. Counselors get a parallel dashboard to monitor the same data at an individual and cohort level, turning a one-time quiz into an ongoing feedback loop between students and counselors.

3.1 Core Modules

- Skill Assessment — slider/questionnaire-based proficiency capture
- Career Matching Engine — profile vs. job-role compatibility scoring
- Skill-Gap Visualization — current vs. required skills per role
- Course Recommendation Engine — gap-to-course mapping
- Counselor Dashboard — radar charts, bulk review, guidance notes
- Skill Growth Tracking — semester-over-semester progress



4. Functional Requirements

FR-1 The system shall provide a skill self-assessment form with proficiency sliders per skill, saved to the student's MongoDB profile document.

FR-2 The system shall compute and display a career-match compatibility percentage for each job role via the Python matching engine.

FR-3 The system shall visualize skill gaps by comparing a student's current skills against the required skills for any selected role.

FR-4 The system shall recommend courses mapped to each identified skill gap, served through the backend API.

FR-5 The system shall provide counselors with a dashboard showing per-student skill radar charts and a bulk cohort view.

FR-6 The system shall let counselors log notes and recommendations directly against a student's profile, persisted in MongoDB.

FR-7 The system shall track and visualize a student's skill growth trend over successive assessments/semesters.

FR-8 The system shall expose recruiter-facing views of skill-verified candidate profiles.

FR-9 The system shall support authenticated, role-based access for students, counselors, recruiters, and training providers.

5. Non-Functional Requirements

Category	Requirement
Usability	Clean, mobile-responsive React + Tailwind UI usable in a single sitting without training.
Performance	Compatibility scores and gap charts render in under 2 seconds for a typical profile.
Scalability	MongoDB schema supports growing numbers of students, roles, and courses without redesign.
Security	Role-based authentication ensures students, counselors, and recruiters see only permitted data.
Reliability	Assessment progress is saved incrementally so no data is lost mid-session.

Category	Requirement
Maintainability	Modular Python API and React components allow independent updates per module.

6. System Workflow

The end-to-end flow connects the student experience to the counselor feedback loop:

- Assess — student completes the skill assessment; responses saved to MongoDB as a structured profile.
- Match — Python engine compares the profile against job_roles data and returns a ranked compatibility percentage per role.
- Visualize — skill-gap chart generated for the student's selected target role.
- Recommend — courses collection queried and mapped to the identified gap skills.
- Review — counselor opens the dashboard, views the radar chart or bulk list, and logs guidance notes.
- Track — skill growth and gap-closure trends update automatically as new assessments are submitted.

7. System Architecture & Tech Stack

7.1 Technology Stack

Layer	Technology
Frontend	React.js, Tailwind CSS, JavaScript (HTML5 / CSS3 foundation)
Backend	Python (REST API layer)
Database	MongoDB (document-based storage for profiles, roles, courses, notes)
Visualization	Chart.js / D3.js (radar charts, skill-gap charts, growth trends)
Design Tools	Figma / Adobe XD (UI prototyping)
Auth & Roles	Role-based access control across student / counselor / recruiter / training-provider accounts

7.2 High-Level Architecture

The React + Tailwind frontend communicates with a Python backend over a REST API. The backend hosts the matching engine and skill-gap logic, and reads/writes all persistent data — student profiles, job roles, courses, and counselor notes — to MongoDB. This separation keeps the assessment UI, the scoring logic, and the data store independently upgradeable.

8. Data Model (MongoDB Collections)

Collection	Purpose
students	Student identity, skill profile, proficiency scores, growth history
job_roles	Role definitions with required skills and proficiency thresholds
courses	Course catalog mapped to specific skills
counselor_notes	Guidance notes and recommendations logged per student
assessments	Historical record of each assessment submission, used for growth tracking

9. Key Modules at a Glance

- Skill Assessment
- Career Matching
- Compatibility Score
- Skill-Gap Visualization
- Course Recommendations
- Counselor Dashboard
- Skill Growth Tracking

10. Uses of This Document

- Guides the development team on exact features to build and their priority.
- Gives evaluators and judges a clear view of scope against the problem statement.
- Helps designers know which screens and charts the UI must cover.

- Supports QA in writing test cases against each functional requirement.
- Assists counselors, recruiters, and training providers in understanding their role in the system.
- Acts as a reference for future maintenance or feature extension.

11. Conclusion

SkillBridge turns vague career advice into a measurable, skill-first process: assess, match, visualize gaps, recommend courses, and track growth — with counselors kept in the loop throughout. Built on a full production-style stack (React, Tailwind CSS, Python, MongoDB), it moves beyond a static prototype into a working portal ready for demonstration and evaluation.